

On Information Collection in Multi-Tagged COTS RFID Systems

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Abstract—We study the problem of target object information collection in multi-tagged COTS RFID systems. Unlike its single-tagged peers, the multi-tagged COTS RFID scenario poses new challenges in devising information collection algorithms: 1) Tags attached to the same object carry identical information. Hence, reusing single-tagged information collection algorithms leads to unnecessary redundancy; 2) Multi-tagged RFID systems are often deployed in applications where tags are vulnerable to damage. Such faulty tags may severely degrade the performance of information collection; 3) Most state-of-the-art information collection algorithms rely heavily on the hashing operation that is not seamlessly supported by the C1G2 standard, rendering these solutions inefficient and impractical. To tackle these technical challenges, this paper makes three contributions. First, we develop an efficient and compact tag pseudo-ID design, enabling the reader to select a single tag from each target object to collect information with only one SELECT command. Second, we construct a robust fault-handling mechanism capable of recognizing faulty tags without executing the entire slot. Third, armed with the above two techniques, we develop a novel information collection algorithm by leveraging the functionality offered by C1G2 to optimize the information collection sequence, thus minimizing the overall execution time. Empirical experiments on a COTS RFID system prototype demonstrate that our algorithm outperforms the best existing solution by 35–50% on average.

Index Terms—Information collection, multi-tagged RFID systems, throughput optimization, C1G2 standard.

I. INTRODUCTION

RADIO frequency identification (RFID) technology [1], [2] exhibits numerous competitive advantages, including non-line-of-sight tracking capability, low-cost deployment, and battery-free operation, rendering it a practicable solution for

a diverse array of Internet of Things (IoT) applications [3], [4] spanning from inventory control [5] to warehouse management [6], behavior recognition [7], and object localization and tracking [8], [9]. An RFID system in its standard form is composed of one reader and a large number of tags that can harvest energy in the RF signal transmitted by the reader to communicate with it in a backscattering manner [9].

In RFID systems, monitoring objects is often hindered by challenging environmental factors such as metals, liquids, extreme temperatures, and strong acids or bases [10], [11], [12]. A practical report [13] reveals that Walmart's RFID applications achieve an object detection rate of 90% at the case level and 95% on conveyor belts, but the rate drops to only 66% within fully loaded pallets due to environmental conditions. In this context, multi-tagged RFID systems, where each object is attached with multiple tags, are emerging as a promising solution to improve the system's fault tolerance and enhance the accuracy and reliability of object monitoring over conventional single-tagged RFID systems [6]. Technically, while a subset of tags on an object becomes faulty due to adverse environmental factors, the remaining non-faulty tags can continue monitoring the object, thereby reducing the risk of data loss or tracking failures. This feature is particularly advantageous in real-world applications such as manufacturing plants with rugged surroundings, where objects are frequently passed through RFID checkpoints, and some of them may be faulty. Attaching multiple tags to each object achieves a persistently high object detection rate [14].

Despite its pivotal role in improving monitoring accuracy and system robustness, the multi-tagged RFID scenario remains largely unexplored in the current body of literature [2], [15], [16], [17], [18], [19]. In this paper, we investigate a practically important problem in multi-tagged commodity off-the-shelf (COTS) RFID systems: how the reader collects information from a given subset of multi-tagged objects in the most time-efficient fashion. This problem arises in various RFID-based IoT applications, such as smart retail [9] and logistics transportation [8], where the real-time and accurate collection of data from objects facilitates precise and reliable decision-making in manufacturing, sales, etc. Unlike its single-tagged peers, the multi-tagged RFID scenario poses three new challenges in devising an object information collection algorithm:

- **Tag redundancy:** As tags attached to the same object carry identical information, one tag's response from each target object is sufficient for information collection. Therefore, we must carefully select a tag from each target object while preventing signal collisions.

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- *Tag faults:* Multi-tagged RFID systems are often deployed in applications where tags are vulnerable to damage [14]. Such faulty tags may severely degrade the performance of information collection. Hence, we need to efficiently identify faulty tags and seek other tags on the same object to collect information.
- *System compatibility:* To be practically implementable, the information collection algorithm should adhere to the Class1 Gen2 (C1G2) protocol. This requirement rules out most state-of-the-art information collection algorithms, which rely on hashing operations that are not efficiently supported by the C1G2 protocol. Currently, Tash is the sole method capable of implementing hashing primitives on COTS RFID systems [20]. However, the necessity for repeated hashing value loading renders it impractical and inefficient.

With the above technical challenges in mind, by looking into existing RFID information collection algorithms proposed over the past decade [1], [2], [3], [4], [5], [19], [21], [22], we find that none of them can tackle the aforementioned challenges well, except the highly inefficient approach by polling each tag sequentially.

Driven by the above observation, in this paper we investigate information collection in multi-tagged COTS RFID systems. To tackle the three challenges, we make three contributions:

- *Efficient tag redundancy elimination:* We develop an efficient and compact tag pseudo-ID design, called tag grouping vector (TGV), allowing the reader to select a tag from each target object with only one SELECT command, while eliminating the interference from other tags without transmitting the entire tag IDs.
- *Robust tag fault handling:* We develop a robust fault-handling mechanism that utilizes the first reply (i.e., RN16) transmitted by a tag at the beginning of a slot (without executing the entire slot) to check if this tag is faulty. Upon recognizing it as faulty, our mechanism promptly terminates the slot and seamlessly continues the collection process with a non-faulty tag.
- *Novel C1G2-compliant algorithm design:* Armed with the above two techniques, we devise a novel information collection algorithm for multi-tagged COTS RFID systems, termed selective polling (SP), by leveraging the functionality offered by C1G2 to optimize the information collection sequence. Our algorithm design strictly adheres to the C1G2 standard without any protocol modification or hardware augmentation. Experiment results validate that our SP algorithm outperforms the best existing solution by an average of 35–50% and also exhibits stronger stability and robustness.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a time-synchronized multi-tagged RFID system composed of a reader, usually connected to a back-end server with powerful computing and storage capabilities, a set $\mathcal{N}$ of N tags, and a set $\mathcal{E}$ of E physical objects, each attached with

n distinct tags, i.e., $N = nE$. Each tag is characterized by a unique $\mathbb{L}$ -bit ID ($\mathbb{L} = 96$ in the current standard) and stores a piece of l -bit object information in its memory. Tags on the same object carry identical l -bit object information, e.g., the storage location of objects. In practice, the l -bit information is changeable and can be updated using an information writing algorithm. We consider a practical situation where a subset $\mathcal{M}$ of tags in $\mathcal{N}$ are faulty. Faulty tags are defined as those that fail to establish reliable communication with the reader. The set $\mathcal{M}$ of faulty tags is unknown to the reader.

We assume that all the tags' IDs are apriori known to the reader. The assumption is justified because (1) the reader can retrieve and store their IDs as soon as they are moved into its coverage area, and (2) the process of storing a tag's ID is particularly fast, typically in the order of milliseconds. Besides, this assumption is a common setting in the RFID information collection problem, broadly used in existing RFID information collection propositions [1], [2], [3], [4], [5], [19], [21], [22], [23].

B. Problem Formulation

We study the target object information collection problem in multi-tagged COTS RFID systems with faulty tags, which is formalized as follows. Given a set $\mathcal{D}$ of D target objects ($\mathcal{D} \subseteq \mathcal{E}$), we seek to devise an information collection algorithm to minimize the expected execution time for gathering the l -bit information from all target objects in $\mathcal{D}$.

This problem seems analogous to the prior tag category/group information collection problem, where tags in the same group can be deemed attached to the same object. Nevertheless, in our problem, since tags on the same object do not share identical category IDs, existing category-level information collection algorithms [3], [21], which rely heavily on these IDs, are inapplicable to our problem. This discrepancy highlights the need for a distinct algorithmic approach tailored to our problem constraints.

For ease of presentation, we refer to the objects whose l -bit information needs to be collected as *target objects*, while the others are called *non-target objects*. Accordingly, the tags attached to target objects are referred to as *target tags*, whereas those attached to non-target objects are called *non-target tags*. Moreover, tags on the same object are referred to as *siblings*.

III. A PRIMER ON C1G2 PROTOCOL

To gain more insights into the design of an information collection algorithm in COTS RFID systems, we introduce technical preliminaries on the C1G2 communication standard.

A. Listen-Before-Talk Rule

The C1G2 protocol specifies air communication between the reader and tags following the “listen-before-talk” rule, in which all the tags strictly follow the commands broadcast by the reader. Specifically, as depicted in Fig. 1, the reader first exploits one or multiple SELECT commands to select a subset of tags to participate in the incoming inventory phase, then broadcasts either a QUERYADJUST or QUERY command carrying parameter

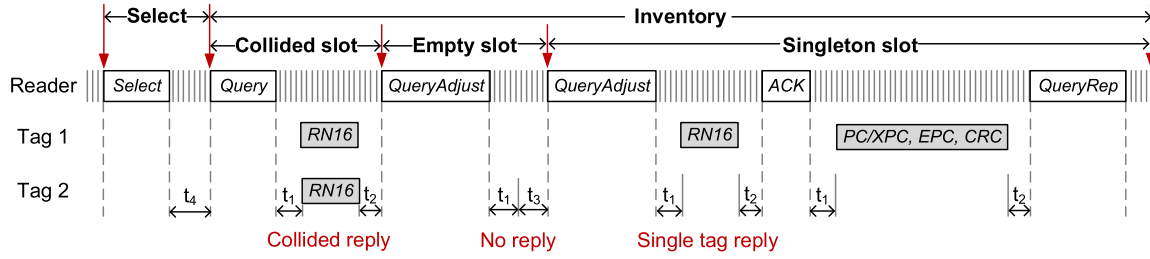


Fig. 1. Illustration of the dialog mechanism between the reader and tags in C1G2 standard.

Q for initiating a frame with $2^Q - 1$ slots. The selected tags randomly choose a slot to reply. With the QUERYADJUST or QUERY command, each selected tag randomly generates a 16-bit random number RN16, and backscatters it to the reader in the selected slot. Upon receiving a legal RN16 from a tag T , the reader acknowledges the RN16 by returning an ACK command to T . Subsequently, T transmits its EPC to the reader. After verifying the received EPC, the reader starts to transact with T in this slot.

In the above protocol, time slots are categorized into *empty*, *singleton*, and *collision* slots: in empty slots, no tag responds; in singleton slots, only one tag responds; in collision slots, multiple tags respond. Notably, only singleton slots can be employed to collect object information, while collision and empty slots are useless.

B. Tag Memory and Select Command

Subsequently, we sketch two pivotal functionalities offered by the C1G2 protocol standard that we will apply in the sequel.

1) *Tag Memory*: C1G2 defines four memory banks for each tag:

- MEMBANK-0 is specified as reserved storage storing the kill and access passwords related to tags;
- MEMBANK-1 is leveraged to store the EPC code;
- MEMBANK-2 stores the tag ID that is loaded by the vendor, and hence unchangeable;
- MEMBANK-3 is a customized area that stores user-defined data. Notably, the size of each memory bank differs in distinct brands or types of COTS RFID tags.

2) *Select Command*: C1G2 strictly specifies that an inventory procedure starts with one or multiple SELECT commands (see Fig. 1), enabling the reader to choose a group of tags to participate in this inventory procedure. Technically, a SELECT command [17] comprises five functional fields: ACTION, MEMBANK, POINTER, LENGTH, MASK.

• MEMBANK, POINTER, LENGTH, and MASK. These four fields collectively specify a bitmask for comparison with the data stored in tag memory. MEMBANK determines which memory bank the bitmask will be compared with. POINTER and LENGTH further specify a starting bit position in MEMBANK to compare with the bitmask, and the length of the bitmask, respectively. MASK then specifies the content of the bitmask to be applied to POINTER and LENGTH.

TABLE I
TAG RESPONSES TO DIFFERENT ACTIONS IN SELECT

Action	Matching tags	Non-matching tags
000	IF $\rightarrow$ A	IF $\rightarrow$ B
001	IF $\rightarrow$ A	do nothing
010	do nothing	IF $\rightarrow$ B
011	A $\rightarrow$ B or B $\rightarrow$ A	do nothing
100	IF $\rightarrow$ B	IF $\rightarrow$ A
101	IF $\rightarrow$ B	do nothing
110	do nothing	IF $\rightarrow$ A
111	do nothing	A $\rightarrow$ B or B $\rightarrow$ A

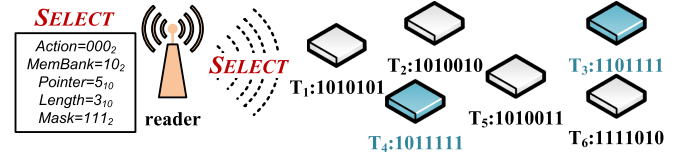


Fig. 2. Illustration of SELECT in C1G2 standard.

• **ACTION**. This field defines an action that is performed by tags. More specifically, with the above four fields, the reader can divide tags into two categories: those tags matching the bitmask specified by the above four fields are called *matching tags*, while the others are termed *non-matching tags*. The reader can thus employ ACTION to impose different actions to matching tags and non-matching tags, i.e., changing their inventoried flags (IFs), as shown in Table I. By setting the IFs of target tags to A or B while the others to the opposite, the reader can select the target tags to participate in the incoming inventory phase while filtering out the non-target tags.

In COTS RFID systems, multiple MASKS fields can be aggregated into one SELECT. For instance, the Alien ALR 9900+ reader [24] allows at most 4 MASKS to be aggregated into one SELECT. By combining the above five fields, we can formalize the SELECT command as

SELECT

$$\left\{ \text{ACTION} \begin{matrix} \text{MEMBANK} \\ 3 \text{ bits}, 2 \text{ bits} \end{matrix}, \left[\begin{matrix} \text{P, L, M} \\ \text{MASK 1} \end{matrix} \right], \left[\begin{matrix} \text{MASK 2} \\ \text{P, L, M} \end{matrix} \right], \left[\begin{matrix} \text{P, L, M} \\ \text{MASK 3} \end{matrix} \right], \dots \right\}.$$

To visually understand the SELECT command, we give an illustrative example, as shown in Fig. 2. To select tags T_3 and T_4 to participate in the upcoming operation, the reader broadcasting

a SELECT with one MASK as $\{000_2, 10_2, [5, 3, 111_2]\}$.¹ On receiving this command, each tag compares the substring starting from the 5th bit with a length of 3 bits in its own MEMBANK-2 (i.e., tag ID) with the MASK 111_2 . With the ACTION 000_2 , the matching tags t_3 and t_4 then set their IFS to $\mathbb{A}$ based on Table I, while the non-matching tags t_1, t_2, t_5, t_6 set their IFS to $\mathbb{B}$. Hence, the reader can select T_3 and T_4 by only allowing the tags with $IF \triangleq \mathbb{A}$ (i.e., T_3 and T_4) to participate in the incoming inventory phase.

IV. INFORMATION COLLECTION ALGORITHM DESIGN: SELECTIVE POLLING

Following the C1G2 standard described in Section III, in this section, we develop a novel information collection algorithm for target multi-tagged objects in COTS RFID systems, termed as selective polling (SP), whose central improvements over existing state-of-the-art solutions are three-fold.

- *Compatible with C1G2*: Different from major existing RFID information collection algorithms [1], [2], [3], [4], [5], [19], [21], [22] relying on either the hashing operation or MAC-layer data acquisition, both of which cannot be efficiently supported in C1G2, our SP algorithm rigorously follows the C1G2 standard without any protocol modification or hardware augmentation, making it seamlessly compatible with COTS RFID systems.
- *Higher throughput*: The traditional polling (TP) method, as the only existing algorithm applicable to our problem, gathers a piece of l -bit object information from a tag by encapsulating its 96-bit original tag ID into a MASK. In comparison, we design a unique pseudo-ID for each tag, called tag grouping vector (TGV), via which our SP algorithm can substantially shorten the length of the MASK to interrogate a tag. Moreover, by combining the ACTION field specified in Table I and the TGV design, our SP algorithm can further decrease the number of SELECTS to collect all the target object's l -bit information, further reducing the overall time overhead.
- *Early termination when reading faulty tags*: To dispose of faulty tags in $\mathcal{M}$, our SP algorithm uses the RN16 backscattered by the selected tag to check if it is faulty, thus detecting faulty tags without the necessity to execute the entire slot. This sharply upgrades both the time efficiency in uplink transmission and the system robustness.

In what follows, we first sketch the design rationale of the SP algorithm, then present the design of the TGV, followed by the design of our SP algorithm.

A. Design Rationale

By combining the C1G2 standard and our problem, we pose three instrumental observations below.

1) *Multi-Tagged Scenario*: Attaching multiple tags to an object can enhance the system's robustness, i.e., if a subset of tags on an object is faulty, the rest tags on this object can continue

monitoring it. Besides, as one tag's response on a multi-tagged object is sufficient to convey its information, we need to select a *non-faulty tag* from each target object as its representative to report its information.

2) *Uplink Transmission*: Since the reader fails to fetch valid information in a collided slot, the pivotal technicality to collect information from tags then lies in avoiding signal interference among them. To further evade empty slots, the information collection operation can only be performed tag by tag. Hence, once encountering a faulty tag, the reader should quickly select one of its siblings as its alternative to report information.

3) *Downlink Transmission*: The key technicality to optimize the time efficiency in the downlink transmission from the reader to tags hinges on decreasing the total number of bits used in SELECT commands, as other commands broadcast by the reader are mandatory, as described in Section III.

With the above three observations, our algorithm follows the polling mechanism, as hashing operations cannot be efficiently implemented on COTS tags and can even lead to empty and collided slots. We carefully devise a unique pseudo-ID (i.e., TGV) for each tag. The TGV is much shorter than the 96-bit original tag IDs, shortening the length of a MASK to interrogate a tag and enabling the reader to select or filter out a set of tags with only one SELECT. Moreover, by combining our TGV design and the ACTION field specified in Table I, our SP algorithm further decreases the number of SELECTS used to collect all target objects' l -bit information. Regarding uplink transmission, our SP algorithm exploits the RN16 transmitted by a selected tag to check if it is faulty. Once recognizing the received RN16 as invalid (i.e., empty or messy code), the reader terminates the slot the tag responds in and promptly selects one of its siblings as its substitution to report information. Hence, the reader can detect faulty tags without executing the entire slot, which sharply upgrades both the time efficiency of uplink transmission and the system's robustness.

B. Pseudo-ID Design and Pre-Load

We construct a unique pseudo-ID for each tag in $\mathcal{N}$, termed as tag grouping vector (TGV). The TGV is pre-loaded into each tag's MEMBANK-3 (i.e., MEMBANK $\triangleq 11_2$), which is a customized block used to store user-defined data, as introduced in Section III. As shown in Fig. 3, the TGV of each tag has two segments: Bucket ID (BID) and Mandatory Tag Code (MTC).

Bucket ID (BID): As the customized block space available in each tag for TGV design is limited, our approach improves the encoding capacity of the TGV by partitioning tags into distinct buckets. This enables tags in different buckets to share the same MTC while being uniquely identified by their respective BIDs. For example, as shown in Fig. 3, if the TGV has a length of 8 bits and contains only the MTC segment, it can represent at most 8 distinct tags, as each tag must be assigned a unique MTC. In contrast, by using the first two bits to group tags into different buckets, the 8-bit TGV can encode up to 16 distinct tags. The design of the MTC will be discussed later; we begin with the structure of the BID.

¹For ease of understanding, in the SELECT command, the POINTER and LENGTH fields are written in decimal, while the others are presented in binary.

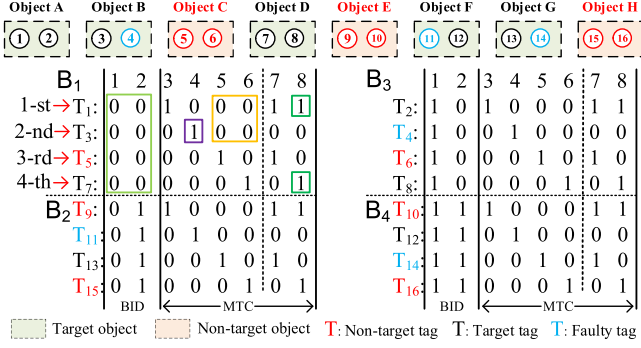


Fig. 3. Illustration of TGV design. 16 tags attached to 8 objects are categorized into 4 buckets.

Specifically, as demonstrated in Fig. 3, we divide tags into a set of disjoint buckets $\{B_i\}_{i=1}^{N/4}$, where each bucket contains 4 tags and any pair of tags in the same bucket are selected from distinct objects. The number of tags per bucket can be adjusted depending on the total number of tags to be encoded; without loss of generality, we set it to 4 in this design. The BID design allows the reader to select a specific bucket or filter out other buckets via a SELECT with only one MASK. For example, as illustrated by the green rectangle in Fig. 3, the reader can select the tag bucket B_1 via $\text{SELECT}\{000_2, 11_2, [1, 2, 00_2]\}$. This is also equivalent to saying that B_2 , B_3 , and B_4 are filtered out by this SELECT.

Mandatory Tag Code (MTC): The primary objective of the MTC design is to minimize the number of MASK/SELECT commands required for tag selection or elimination, thereby reducing the overall time overhead. A straightforward approach is to encode four tags within each bucket using two bits. However, this strategy does not guarantee that any subset of tags in the same bucket can be selected using a single MASK, as the 0/1 distributions in each bit column can not cover all possible tag combinations. To address this limitation, we design the MTC with two complementary components.

The rationale behind the first component is to construct a sequence of contiguous 0 s in each column of the TGVs, thereby enabling the reader to select or filter out a group of contiguous tags within a bucket using a single SELECT command. As depicted in Fig. 3, since each bucket contains 4 tags, the first component is assigned the leftmost 4 bits of the MTC. For the i th tag ($1 \leq i \leq 4$) in a bucket B_k , the i th bit within the first component is configured as 1 while others are set to 0. For instance, in bucket B_1 , the first component for the 2-nd tag T_3 is set to “0100”. Such design settings enable the reader to select or filter out a single tag or a set of contiguous tags via a SELECT command with two MASKS. For instance, to select T_3 , the reader can issue a SELECT with two MASKS as $\{000_2, 11_2, [1, 2, 00_2], [4, 1, 1_2]\}$, as depicted by the green and purple rectangles. Similarly, the reader can select a set of contiguous tags (e.g., the 1-st tag T_1 and the 2-nd tag T_3 in B_1) with $\text{SELECT}\{000_2, 11_2, [1, 2, 00_2], [5, 2, 00_2]\}$, as depicted by the green and orange rectangles in Fig. 3.

The rationale behind the second component is to construct an alternate distribution between 0 and 1 in each column of

the TGVs, thereby enabling the selection or filtering of non-contiguous (i.e., discrete) tags within a bucket using a single SELECT command. This component occupies the rightmost 2 bits of the MTC. In any bucket, the second component of the 1-st, 2-nd, 3-rd, and 4-th tags are configured as “11”, “00”, “10”, and “01” respectively, as visualized in Fig. 3. Such design settings allow the reader to select or filter out any subset of discrete tags in the same bucket via a SELECT command with two MASKS. For instance, to select two discrete tags in B_1 (e.g., the 1-st tag T_1 and the 4-th tag T_7), the reader can issue a SELECT as $\{000_2, 11_2, [1, 2, 00_2], [8, 1, 1_2]\}$, as depicted by the green rectangle and the two blue boxes in Fig. 3.

To this end, by combining these two components, the MTC allows the reader to flexibly select or eliminate any subset of tags within a bucket using only one SELECT command.

C. SP Algorithm Design

Before executing our algorithm, we pre-load the corresponding TGV to each tag. The pre-loading operation is widely used in existing tag monitoring algorithms [6], [23] in COTS RFID systems. It is also supported by mainstream COTS tags, such as Impinj [25] and Alien [24] tags. In major cases, the TGV is created and then pre-loaded into tags by users. Alternatively, users can ask the RFID manufacturer to write the TGV into tags’ user memory while manufacturing them [24], [25]. In either case, as the TGV design can be reused, the pre-loading operation only needs to be performed once. The same TGV can be reused in the subsequent information collection operations. Hence, the cost related to loading TGVs is amortized, even in dynamic environments where tags are frequently added or removed.

Armed with the TGV, our SP algorithm further decreases the overall time overhead in collecting the l -bit information of target objects in $\mathcal{D}$, by leveraging the ACTION field specified in Table I. As tags on the same object carry identical object information, to collect all the target objects’ information, the reader only needs to interact with a subset G of the buckets—specifically, half of the total number of buckets—to collect the l -bit information of all target objects. For instance, in Fig. 3, to collect all target objects’ information, it suffices for the reader to transact with only buckets B_1 and B_2 . Moreover, because faulty tags in the system are unknown in advance, tags in the remaining buckets B_3 and B_4 serve as backup substitutes during the collection process. Our SP algorithm processes the information collection bucket by bucket in G . For any bucket in G , such as bucket B_2 in Fig. 3, the reader collects the l -bit information of target objects involved in B_2 through the following two steps.

Step 1. Selecting target tags in B_2 : As shown in Fig. 3, objects A, B, D, F, and G are target objects, while the others are non-target. In bucket B_2 , tag T_{11} can be used to collect the l -bit information of target object F, while tag T_{13} can be used for target object G. Let $\mathcal{H}_2$ and $\mathcal{R}_2$ denote the sets of target and non-target tags in bucket B_2 , respectively. As depicted in Fig. 4, we have $\mathcal{H}_2 = \{T_{11}, T_{13}\}$ and $\mathcal{R}_2 = \{T_9, T_{15}\}$, where T_{11} is a faulty tag. Since only tags with $\text{IF} \triangleq \mathbb{A}$ are permitted to participate in the subsequent information collection phase (i.e., Step 2), we assign $\text{IF} \triangleq \mathbb{A}$ to the target tags in $\mathcal{H}_2 = \{T_{13}, T_{11}\}$,

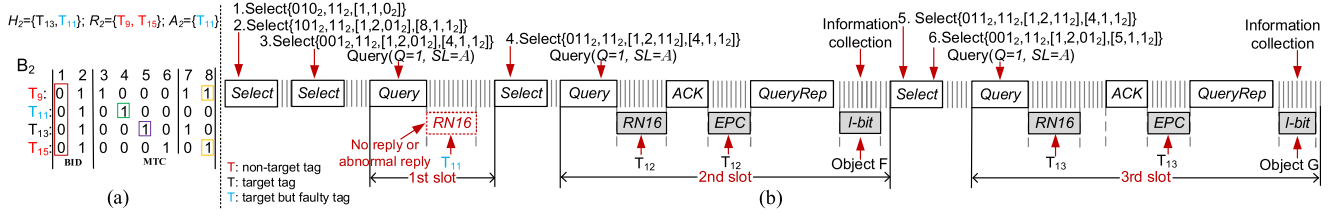


Fig. 4. (a) Pseudo-IDs for B_2 in Fig. 3; (b) Procedure of target object information collection based on (a).

TABLE II
FUNCTIONALITIES OF DISTINCT SELECTS USED IN FIG. 4(b)

SELECT	Matching tags	Non-matching tags
1	do nothing	$B_3, B_4 \rightarrow \mathbb{B}$
2	$R_2 \rightarrow \mathbb{B}$	do nothing
3	$T_{11} \rightarrow \mathbb{A}$	do nothing
4	$T_{12} \rightarrow \mathbb{A}$	do nothing
4	$T_{12} \rightarrow \mathbb{B}$	do nothing
6	$T_{13} \rightarrow \mathbb{A}$	do nothing

while setting the IFS of all other tags in the system to either NULL or $\mathbb{B}$.

To do this, the reader first configures the IFS of all tags in B_3 and B_4 as $\mathbb{B}$, by issuing a SELECT with a MASK targeting the first bit of the BID as $\text{SELECT}\{010_2, 11_2, [1, 1, 0_2]\}$, as depicted by the 1-st SELECT in Fig. 4(b). Upon receiving this command, non-matching tags (i.e., tags in B_3 and B_4) set their IFS to $\mathbb{B}$ based on the ACTION 010 specified in Table I, while matching tags (i.e., tags in B_1 and B_2) do nothing to their IFS. The functionalities of all SELECTS used in Fig. 4(b) are detailed in Table II. Subsequently, the reader filters out non-target tags in $\mathcal{R}_2 = \{T_9, T_{15}\}$ by setting their IFS to $\mathbb{B}$ via $\text{SELECT}\{101_2, 11_2, [1, 2, 0_{12}], [8, 1, 1_2]\}$, i.e., the 2-nd SELECT in Fig. 4(b). As a result, tags fall into three categories:

- With $\text{IF} \triangleq \mathbb{B}$, tags in B_3 , B_4 , and $\mathcal{R}_2$ are filtered out.
- The IFS of tags in B_1 have not been set yet (i.e., NULL), and their IFS will remain NULL until the reader initiates transactions with them.
- The IFS of tags in $\mathcal{H}_2$ are also NULL at this stage. Their IFS will be set to $\mathbb{A}$ in the next step to enable participation in the upcoming information collection round.

Step 2. Collecting target object information: The reader collects the l -bit object information from each tag in $\mathcal{H}_2$ by setting its IF to $\mathbb{A}$ and assigning it to a dedicated slot. In each slot, the reader verifies the tag's status by examining the RN16 it transmits. If the received RN16 is invalid (e.g., empty or undecodable), the reader identifies the tag as faulty, terminates the slot, and selects one of its sibling tags as a substitute to report the l -bit information. Otherwise, if the RN16 is valid, the slot is executed successfully, and the reader collects the corresponding object's l -bit information.

Concretely, as shown in Fig. 4(b), to collect the l -bit information of target object F from its attached tag T_{11} , the reader broadcasts the 3-rd $\text{SELECT}\{001_2, 11_2, [1, 2, 0_{12}], [4, 1, 1_2]\}$ to all tags in the system. This SELECT sets the IF of T_{11} to

$\mathbb{A}$ while leaving all other tags unaffected. Subsequently, the reader initiates an inventory round consisting of a single slot (i.e., $2^Q - 1 = 1$) by issuing a QUERY command with $Q = 1$ and the selected flag (SL) set to $\mathbb{A}$.² This ensures that only tags with $\text{IF} \triangleq \mathbb{A}$ (i.e., T_{11}) are eligible to respond. However, as T_{11} is a faulty tag, it fails to backscatter a valid RN16. The reader detects this failure, terminates the slot, and selects one of its sibling tags (e.g., T_{12}) as a substitute to report the l -bit information of object F. This is achieved by broadcasting the 4-th $\text{SELECT}\{011_2, 11_2, [1, 2, 11_2], [4, 1, 1_2]\}$, as illustrated in Fig. 4(b). This command sets the IF of T_{12} to $\mathbb{A}$ without affecting other tags. The reader then successfully collects the l -bit information of object F from T_{12} in the 2-nd slot. Similarly, after resetting T_{12} 's IF to $\mathbb{B}$ via the 5-th SELECT, the reader can collect the l -bit information of target object G from T_{13} by using the 6-th $\text{SELECT}\{001_2, 11_2, [1, 2, 0_{12}], [5, 1, 1_2]\}$ to setting its IF to $\mathbb{A}$ in the 3-rd slot.

After executing the above procedure, the reader shifts the information collection operation from B_2 to B_1 , and the operation is repeated iteratively for each bucket in the selected subset G until all target objects' l -bit information is collected.

V. THEORETICAL PERFORMANCE ANALYSIS

In this section, we evaluate the theoretical performance of our SP algorithm in terms of the expected execution time.

Technically, as the execution time of an RFID information collection algorithm can be measured by the total number of communication bits χ used in this algorithm, we then need to compute $\mathbb{E}[\chi]$ for our SP algorithm. We denote by χ_v and χ_s the numbers of bits exploited in the inventory and select phases under our SP algorithm, respectively, and by m the number of faulty tags attached to an arbitrary target object.

Our analysis comprises three steps: First, we analyze the length of each command in C1G2. Second, we compute $\mathbb{E}[\chi_v]$ by calculating the expected number of bits for collecting l -bit information of F in three cases of m and then generalizing to all D objects in $\mathcal{D}$. Third, we derive $\mathbb{E}[\chi_s]$ using a similar method as for $\mathbb{E}[\chi_v]$, and combine it with $\mathbb{E}[\chi_v]$ to obtain $\mathbb{E}[\chi]$.

A. Time Overhead of Commands

For ease of presentation, we refer to the inventories (slots) in which faulty and non-faulty tags participate as *truncated* and *complete inventories (slots)*, respectively. According to the

²Tags with $\text{IF} \triangleq \text{SL}$ are allowed to attend the upcoming inventory round.

TABLE III
LENGTHS OF COMMANDS/REPLIES IN THE INVENTORY PHASE

Operation	Command/reply	Length (bits)
Information collection	REQ_RN	40
	READ	58
	GRC	16
	Object information	l
Others	QUERY	22
	RN16	16
	QUERYADJUST	9
	ACK	18
	NAK	8
	EPC+GRC+PC	128
	QUERYREP	4

C1G2, the length of each command or parameter used in our algorithm is standardized as follows.

1) *Select Phase*: In a SELECT, those mandatory fields take a total of 29 bits, while the optional fields, including POINTER, LENGTH, and MASK are customized and hence their lengths are variable.

2) *Inventory Phase*: The length of each command/reply in the inventory phase is demonstrated in Table III.

With the above specification in C1G2, in our SP algorithm, a truncated inventory consumes a total of 38 bits, whereas a complete inventory consumes $(302 + l)$ bits. Besides, armed with the TGV design, our SP algorithm requires at most $2(\log n + \log E) + 37 = 2 \log N + 37$ bits to issue a SELECT, and it takes one SELECT (i.e., at most $(2 \log N + 37)$ bits) to initiate a truncated inventory while using two SELECTS (i.e., up to $(4 \log N + 74)$ bits) to initiate a complete inventory.

B. Performance Analysis for SP Algorithm

With the above analysis, we derive the Lemma 1 below.

Lemma 1: In our SP algorithm, the expected time overhead for collecting the l -bit information of an arbitrary object F in the inventory phase, denoted by $\mathbb{E}[\chi_v^F]$, is given as follows:

$$\mathbb{E}[\chi_v^F] = \begin{cases} 302 + l & m = 0 \\ \sum_{p=1}^{m+1} \Delta(n, m, p) \cdot (264 + l + 38p) & 1 \leq m \leq n-1 \\ 38n & m = n, \end{cases}$$

where $\Delta(n, m, p) = \frac{n-m}{n-p+1} \prod_{i=1}^{p-1} \frac{m-i+1}{n-i+1}$.

Proof: The above three cases are analyzed as follows.

Case 1: $m = 0$. The reader can collect the l -bit information of F with only one complete inventory, which incurs an overall time overhead of $(302 + l)$ bits.

Case 2: $m = n$. The reader, even interrogating all the n faulty tags on F, still cannot acquire the l -bit information of F, which consumes a communication overhead of $38n$ bits in the inventory phase.

Case 3: $1 \leq m \leq n-1$. We denote by P a random variable representing the number of slots employed to collect the l -bit object information of F, among which the first $P-1$ ones are truncated slots and the last one is a complete slot. This case

corresponds to an overall overhead of

$$(P-1)38 + 302 + l = 264 + l + 38P \quad (1)$$

bits in the inventory phase. We next analyze the probability $\Pr[P = p]$. $P = p$ indicates that our SP algorithm successfully collects F's information using p slots, where each of the first $p-1$ slots is a truncated slot involving a faulty tag, and the p th slot is a complete slot involving a non-faulty tag. In our SP algorithm, once a tag is identified as faulty in a truncated slot, it is eliminated from all subsequent slots. Given that F is attached with m faulty tags and $n-m$ non-faulty tags, the probability that the i th slot among the first $p-1$ slots involves a faulty tag is $\binom{m-i+1}{1} / \binom{n-i+1}{1}$. The probability that the p th slot involves a non-faulty tag is $\binom{n-m}{1} / \binom{n-p+1}{1}$. Hence, $\Pr[P = p]$ can be calculated as

$$\Pr[P = p] = \frac{\binom{n-m}{1}}{\binom{n-p+1}{1}} \prod_{i=1}^{p-1} \frac{\binom{m-i+1}{1}}{\binom{n-i+1}{1}} = \Delta(n, m, p), \quad (2)$$

where $1 \leq p \leq m+1$. It follows from (1) and (2) that

$$\mathbb{E}[\chi_v^F] = \sum_{p=1}^{m+1} \Delta(n, m, p) \cdot (264 + l + 38p).$$

Combining the three-fold analysis leads to Lemma 1. ■

With Lemma 1, we can derive $\mathbb{E}[\chi_v]$, as stated in Lemma 2.

Lemma 2: $\mathbb{E}[\chi_v] = \frac{D}{2^n} \{302 + l + 38n + \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \binom{n}{m} \cdot \Delta(n, m, p)(264 + l + 38p)\}$.

Proof: Let M denote another random variable representing the number of faulty tags attached to F. Assuming that tags in the system are distributed uniformly among $\mathcal{N}$, each having the same probability (i.e., $1/2$) to be non-faulty or faulty, we yield

$$\Pr[M = m] = \binom{n}{m} \left(\frac{1}{2}\right)^m \left(\frac{1}{2}\right)^{n-m} = \binom{n}{m} \frac{1}{2^n}, \quad (3)$$

where $0 \leq m \leq n$.

Given that **Case 1** and **Case 2** involve only one value for m , the crux of the proof shifts to **Case 3** with $1 \leq m \leq n-1$. Intuitively, by (2) and (3), probability $\Pr[P = p]$, conditioned by $M = m$, can be calculated as

$$\begin{aligned} \Pr[P = p | M]_{p,m} &= \Pr[M = m] \cdot \frac{n-m}{n-p+1} \prod_{i=1}^{p-1} \frac{m-i+1}{n-i+1} \\ &= \binom{n}{m} \frac{1}{2^n} \cdot \Delta(n, m, p), \end{aligned} \quad (4)$$

where $1 \leq p \leq m+1$.

By Lemma 1, we obtain that the expected time overhead for gathering the l -bit information of object F in the inventory phase, denoted by $\mathbb{E}[\mathcal{U}]$, is

$$\begin{aligned} \mathbb{E}[\mathcal{U}] &= \Pr[M = 0] \cdot (302 + l) + \Pr[M = n] \cdot 38n \\ &\quad + \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \Pr[P = p | M]_{p,m} \cdot (264 + l + 38p). \end{aligned}$$

By generalizing F to all the D target objects in $\mathcal{D}$, we yield $\mathbb{E}[\chi_v] = D \cdot \mathbb{E}[\mathcal{U}]$. The Lemma 2 is thus proved. ■

Subsequently, with Lemma 2, we can further compute the expected execution time of our SP algorithm, denoted by $\mathbb{E}[\chi]$, as stated in Theorem 1.

Theorem 1: The expected execution time of the SP algorithm can be denoted by

$$\mathbb{E}[\chi] = \frac{D}{2^n} \left\{ 2 \log N(n+2) + 376 + 75n + l + \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \cdot \right. \\ \left. \binom{n}{m} \cdot \Delta(n, m, p) \cdot (2 \log N(p+1) + 75P + l + 301) \right\} \\ + \frac{(2 \log N + 37)(N + 4n)}{4n}.$$

Proof: As described previously, our SP algorithm takes $2(\log n + \log E) + 37$ and $4(\log n + \log E) + 74$ bits in the select operation to initiate a truncated inventory and a complete inventory, respectively. Moreover, it uses a SELECT to exclude tags not belonging to G_i and another $E/4$ SELECTS to filter out non-target tags in all the $E/4$ buckets in G_i . By Lemma 2 and $E = N/n$, our SP algorithm then takes a total of

$$\mathbb{E}[\chi_s] = D \cdot \left\{ \Pr[M = 0] \cdot (4 \log N + 74) + \Pr[M = n] \cdot \right. \\ \left. n(2 \log N + 37) + \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \Pr[P | M]_{p,m} \cdot (p+1) \cdot \right. \\ \left. (2 \log N + 37) \right\} + \frac{(2 \log N + 37)(N + 4n)}{4n} \quad (5)$$

bits in the select phase to collect the l -bit information of all the D target objects in $\mathcal{D}$. As $\mathbb{E}[\chi] = \mathbb{E}[\chi_s] + \mathbb{E}[\chi_v]$, combining Lemma 2 and (5) then leads to Theorem 1. ■

C. Theoretical Performance Comparison

We propose an enhanced version of the TP method, termed as enhanced TP (ETP), as a baseline for performance comparison. It first selects a non-faulty tag from each target object, then performs the information collection process by querying the selected tags based on their 96-bit original IDs.

Theorem 2: SP outperforms ETP by at least 40% in terms of the expected execution time.

Proof: Similarly, we can use the same analytical method as for our algorithm to calculate the expected execution time of the ETP algorithm. Specifically, as it needs to execute the entire time slot to identify a faulty tag, given an arbitrary object F , $\mathbb{E}[\chi_v^F]$ under the ETP algorithm is represented as follows:

$$\mathbb{E}[\chi_v^F] = \begin{cases} 302 + l & m = 0 \\ \sum_{p=1}^{m+1} \Delta(n, m, p) \cdot (302p + l) & 1 \leq m \leq n-1 \\ 302n & m = n. \end{cases}$$

According to the similar analysis as for Lemma 2, we further derive $\mathbb{E}[\chi_v]$ in the ETP algorithm as

$$\mathbb{E}[\chi_v] = \Pr[M = 0] \cdot (302 + l) + \Pr[M = n] \cdot 302n$$

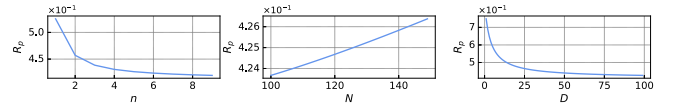


Fig. 5. Variation trend of R_p w.r.t various parameters.

$$+ \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \Pr[P | M]_{p,m} \cdot (l + 302p). \quad (6)$$

Besides, since the ETP algorithm interrogates tags by broadcasting their 96-bit original IDs, it takes roughly 125 bits to complete an inventory phase. Consequently, by (5), $\mathbb{E}[\chi_s]$ in the ETP algorithm can be expressed as

$$\mathbb{E}[\chi_s] = \frac{125N}{4n} + D \cdot \left\{ \Pr[M = 0] \cdot 250 + \Pr[M = n] \cdot 125n \right. \\ \left. + \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \Pr[P | M]_{p,m} \cdot (p+1) \cdot 125 \right\}. \quad (7)$$

By combining (6) and (7), we can obtain the expected execution time of the ETP algorithm, denoted by $\mathbb{E}[\hat{\chi}]$, as follows:

$$\mathbb{E}[\hat{\chi}] = \frac{D}{2^n} \left\{ \sum_{m=1}^{n-1} \sum_{p=1}^{m+1} \binom{n}{m} \cdot \Delta(n, m, p) \cdot (427p + 125 + l) \right. \\ \left. + l + 552 + 427n + \right\} + \frac{125N}{n}. \quad (8)$$

To evaluate the performance improvement of our SP algorithm over the ETP algorithm, we define the following equation: $R_p = 1 - \mathbb{E}[\chi]/\mathbb{E}[\hat{\chi}]$. Fig. 5 illustrates the variation trend of R_p regarding n , N , and D . We observe from Fig. 5 that the value of R_p consistently exceeds 0.4 under distinct parameter settings. Hence, the proof is completed. ■

VI. IMPLEMENTATION & EVALUATION

In this section, we implement our algorithm SP using the COTS RFID reader and tag, and evaluate its empirical performance compared to state-of-the-art solutions.

A. Experimental Methodology

Reader configurations: Our experiments take Impinj R700 as the RFID reader applied [25], which is the latest Enterprise-class reader manufactured by Impinj. It features powerful communication capability and high capabilities, offering a best-in-class C1G2 RFID solution for upper-level applications. The Impinj R700 is connected to the HAUHF-2599, a circularly polarized antenna carrying 9 DBi information gain and operating at 902-928 MHz frequency band, enabling users to implement a complete information collection cycle through bit-level masks. Additionally, the Impinj R700 supports the aggregation of multiple MASKS in a single SELECT command.

Tag settings: As described in Section IV, the pseudo-ID pre-loading operation is supported by all mainstream COTS RFID

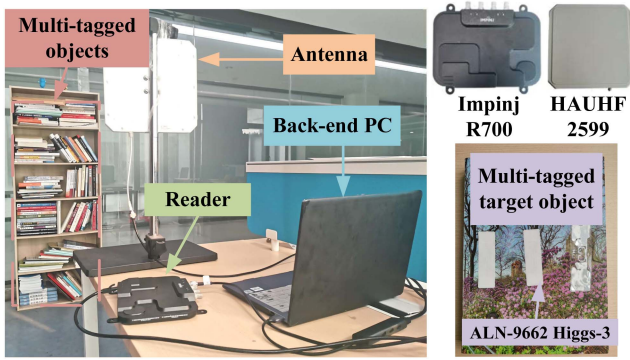


Fig. 6. Experimental scenario.

tags, e.g., Impinj [25] and Alien [24] tags. The time overhead in this operation depends on the space size of user-customizable memory (i.e., MEMBANK-3) offered by various tags. In our experiments, we employ the ALN-9662 Higgs-3 tag [24] to evaluate the algorithm performance, which provides users with a 512-bit MEMBANK-3, of which the first 256 bits are applied in our experiments. Specifically, the first 128 bits are used to store the TGV design in our SP algorithm, while the remaining 128 bits are for encoding the l -bit object information. Our system has a total of 600 tags, i.e., $N = 600$.

Multi-tagged objects: As illustrated in Fig. 6, there are a total of 60 (i.e., $E = 60$) books in our system, among which D ($10 \leq D \leq 50$) ones are regarded as target objects, while the rest $60 - D$ ones are non-target objects. Besides, each book is attached with n ($1 \leq n \leq 10$) tags, among which m ($1 \leq m < n$) ones are considered as faulty tags.

Benchmark algorithms: We select two state-of-the-art algorithms as baselines for performance evaluation, including Single-hash Information Collection (SIC) [15] and Multi-hash Information Collection (MIC) [15]. To mitigate tag collisions, SIC performs hashing operations sequentially on tags, while MIC conducts multiple hashing operations simultaneously for them. By exploiting the Tash framework [20] to enable the hashing primitive in C1G2, we implement the core designs of both SIC and MIC in our system prototype. In addition, we introduce two other baseline algorithms for performance comparison. The first one is an enhanced version of the traditional polling method, termed Enhanced Traditional Polling (ETP) [26]. ETP sequentially polls tags attached to the same target object until a non-faulty tag is identified. The second one, termed Category Information Collection (CIC), is inspired by the key idea in [27], which regards the set of tags attached to the same object as a tag category and samples one non-faulty tag from each category via hashing operations.

Experimental scenario: As illustrated in Fig. 6, our experimental setup emulates an RFID-enabled library system, where books attached with multiple tags are densely arranged within a cabinet measuring 1.6 m in height. Tags are attached either to the horizontal or vertical surfaces of the books. The cabinet is placed at a fixed distance of 2 m from the antenna to eliminate multipath effects. The antenna is initially positioned at a height of 120 cm,

with both the horizontal and vertical angles between the antenna and the cabinet initialized to 0° . All equipment remains stationary throughout the experiment. We conduct 20 experimental runs to compare the performance of different algorithms.

B. Experimental Results

We first evaluate the performance comparison among various algorithms, followed by an analysis of the impact of environmental factors on the performance of our SP algorithm.

Performance comparison on S : We denote by S the number of SELECT commands used in each algorithm. From Fig. 7(a), it is noticeable that with m varying from 1 to 9, S in all the algorithms first slightly increases and then dramatically goes up. This trend is attributed to the fact that an increase in the number of faulty tags results in an increase in the number of SELECT commands required. In Fig. 7(b), as n increases, S in both SIC and MIC constantly increases, while that in the three other algorithms exhibits a gradual decline. This is due to that the growth in n results in a greater probability of selecting a non-faulty tag from each target object for ETP, CIC, and SP. In Fig. 7(c), S in all the algorithms is linearly increasing in D , because an increase in D necessitates the execution of an expanded array of information collection tasks. Besides, as the SELECT command is used only in the SELECT phase, S is then independent of l , as illustrated in Fig. 7(d).

In comparison, due to their strategy of querying all tags in the system, SIC and MIC prominently lag behind the three other algorithms in terms of S . CIC slightly outperforms ETP, as ETP polls each tag by querying its relatively long original IDs, whereas CIC samples tags employing shorter hash values. Our algorithm SP consistently outperforms CIC by 35-50% in terms of S . For instance, as shown in Fig. 7(a), when m increases from 1 to 9, S in CIC grows from 350 to 880, while that in SP fluctuates between 140 and 580. On average, SP reduces S by 41.4% compared to CIC. Similarly, in Fig. 7(c), as D increases from 10 to 45, SP achieves an average reduction of 35.23% in S over CIC.

Performance comparison on χ : Fig. 8 shows the performance comparison among various algorithms on the number χ of bits exchanged between the reader and tags. In general, the variation trends of χ in terms of m , n , and D are similar to that of S regarding m , n , and D , respectively. Specifically, χ in SP is proportional to D , and its slope slightly increases in m while modestly decreasing in n . These variation trends are consistent with our theoretical performance analysis. Besides, from Fig. 8(d), it can be observed that χ in each algorithm is modestly increasing in l . This is because the growth in l leads to a corresponding increase in the amount of information that needs to be collected by each algorithm.

Similar to Figs. 7 and 8 shows that SP is dramatically superior to the four other algorithms, showcasing higher scalability, stability, and robustness. For instance, in Fig. 8(b), with n varying from 3 to 9, χ in SP decreases from 55000 to 45000, whereas that in CIC declines from 100000 to 56000. In comparison, SP not only reduces χ by at least 35.8% on expectation but also undergoes a smaller performance variation. A similar trend can

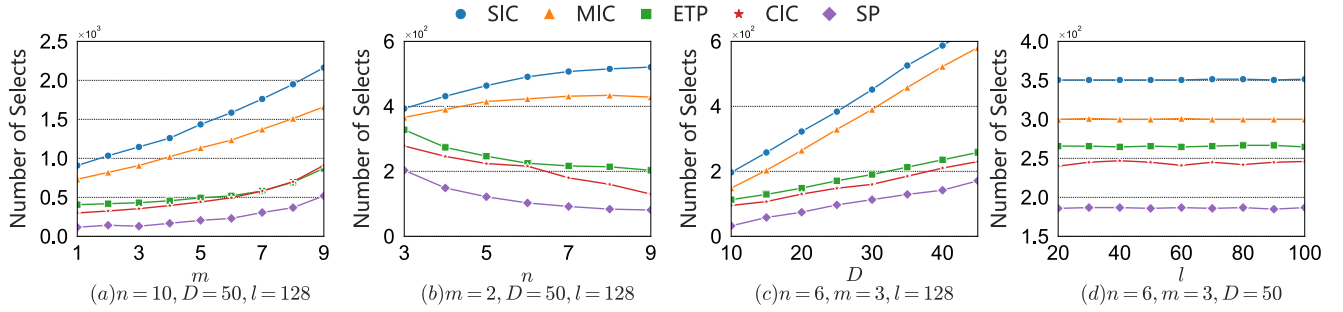


Fig. 7. Performance comparison among various algorithms on the number S of SELECT commands required.

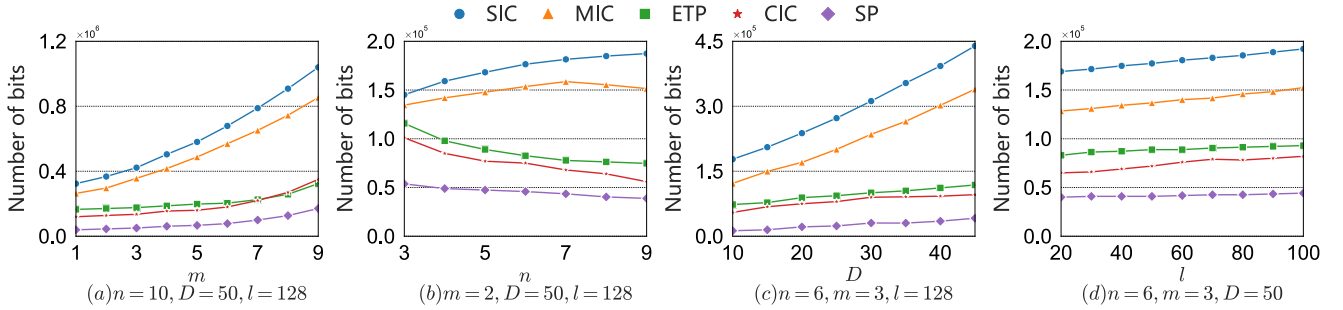


Fig. 8. Performance comparison among various algorithms on the number χ of bits exchanged between the reader and tags.

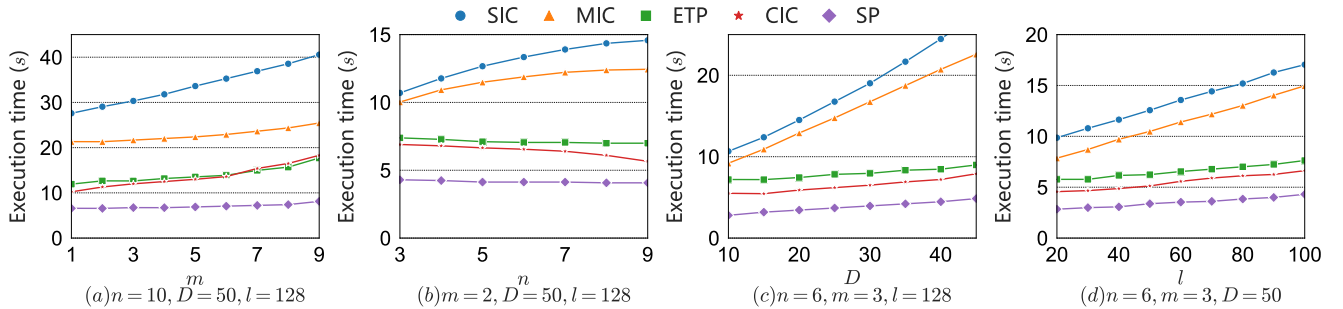


Fig. 9. Performance comparison among various algorithms on the execution time Λ .

also be observed from Fig 8(d), where SP outperforms the four other algorithms by at least 40% in χ . In particular, Fig. 8(a) demonstrates a substantial enhancement in the performance gap between SP and CIC as m ranges from 1 to 9. It reaches as large as 850000 at $m = 9$. This indicates that SP exhibits higher fault tolerance than the four other algorithms.

Performance comparison on Λ : Let Λ denote the aggregate execution time of each algorithm. Notably, the unit of Λ is a second. By comparing Figs. 9 and 8, we observe that Λ generally follows similar variation trends as χ , but it is less sensitive to these four parameters than χ . This is because Λ encompasses not only χ but also other communication overhead, e.g., the interval duration between slots. Similar to Figs. 7, 8, and 9 also demonstrates that SP significantly upgrades the time efficiency, performance stability, and robustness over SIC, MIC, CIC, and

ETP. For instance, as shown in Fig. 9(b), with n increasing from 3 to 9, Λ in SP undergoes a performance fluctuation of 4.2%, while that in SIC, MIC, CIC, and ETP exhibits performance variations of 26%, 15%, 10.3%, and 6.6%, respectively. Compared to the four algorithms, SP reduces Λ by at least 38.5% on expectation.

Impact of environmental factors: We next analyze the impact of environmental factors on the performance of our SP algorithm. As shown in Fig. 10(a), the execution time Λ of SP grows slightly as the reader's emission peak power decreases from -10 to -35 dBm. In particular, it remains relatively stable when the emission power is above -25 dBm. Similarly, the performance of the algorithm remains consistent as the distance between the reader and tags ranges from 2 to 8 m. Moreover, Fig. 10(b) illustrates that the performance of SP does not vary significantly across different heights of the antenna. Fig. 10(c) presents the

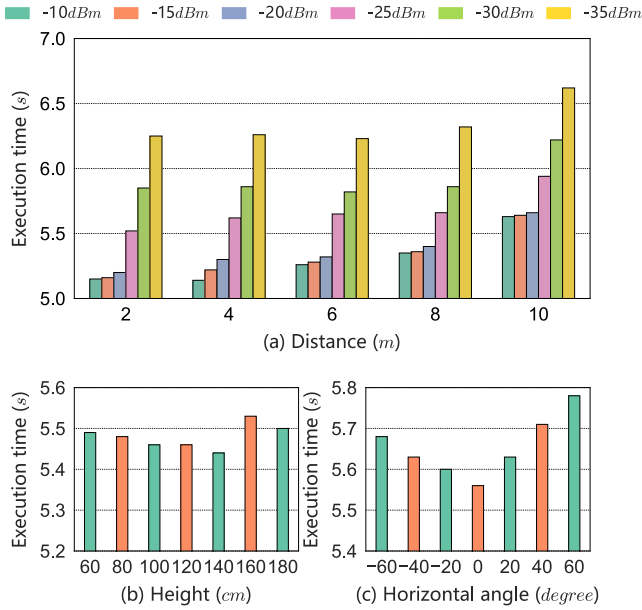


Fig. 10. The impact of environmental factors on our SP algorithm.

results under different horizontal angles between the antenna and the cabinet. Here, we observe that Λ in SP exhibits a symmetrical trend with respect to the angle, reaching its minimum at 0° . This trend is likely due to the influence of tag orientation on the link angle and polarization, resulting in phase changes.

Overall, we can draw the following conclusions from our experimental results: (1) SIC, MIC, CIC, and ETP are more sensitive with m , n , D , and l than SP; (2) S , χ , and Λ in SP are convex in m and n , while χ and Λ in SP linearly increase in D and l ; (3) our algorithm SP outperforms all the involved algorithms with non-negligible performance gains in terms of time overhead, stability, and robustness, achieving an average performance improvement of 35-50% over CIC.

VII. RELATED WORKS

Information collection is one of the most fundamental tasks in RFID systems. State-of-the-art RFID information collection algorithms can be broadly divided into three categories.

A. Information Collection on All Tags

This category focuses on retrieving information from all tags in the system. [15] designed the Multi-Hash Information Collection (MIC) algorithm that collects tag information in the single-reader scenario via multiple hashing operations. [16] then extended this problem to multi-reader scenarios with the Bloom Filter-based Information Collection (BIC) algorithm, which uses a distributively constructed bloom filter to mitigate tag collisions. Subsequent research aimed at decreasing energy consumption and time overhead. Specifically, [18] introduced two energy-efficient polling-based protocols, termed as tag ordering polling (TOP) and enhanced TOP (ETOP). Technically, TOP employs multiple carefully designed vectors to indicate which tags should respond in which slots, while

ETOP further decreases the number of vectors exploited in TOP. Additionally, [19] developed the incremental polling protocol (IPP) that exploits polling vectors to reduce the time overhead in interrogating tags. [2] designed the partial double polling-based (PDP) algorithm that can simultaneously query a pair of tags via so-called double indexes.

B. Information Collection on Target Tags

In contrast, target-tag information collection concentrates on collecting information from a predefined set of tags rather than all tags in the system. To address this issue, [5] developed the minimal perfect hashing-based information collection (PIC) algorithm. PIC collects information from target tags by utilizing a compact data structure to filter out non-target tags. [4] then extended this problem to multi-reader scenarios by designing the target tag information collection (TIC) algorithm. Furthermore, to enhance the system's fault tolerance, [22] and [1] delved into the problem of target tag information collection with anomalies. [22] augmented the BIC algorithm to collect sensory data from unknown target tags, while [1] proposed the multi-framed hierarchical-hashing data collection (MHDC) algorithm to handle the presence of unexpected tags.

C. Category-Level Information Collection

Tags belonging to the same category or deployed in the same physical environment can collectively constitute a category or group. Within a group, each tag stores an identical group ID and carries identical category-level information. To gather such information, [3] proposed two algorithms: two-phase sampling (TPS) and back-and-forth sampling (BFS). TPS first adopts the group ID to locate a specific group, then selects a single tag from that group to report the category-level information. For further enhancement, BFS utilizes both a step-forward frame and a step-backward parsing scheme to collect such information without prior knowledge of tag IDs and group IDs. The geographical correlation-based RF-data collection (GRC) algorithm [21] gathers category-level sensory data by grasping geographical correlations among tags, i.e., data sensed by nearby sensor-augmented tags exhibits similarity.

VIII. CONCLUSION

In this paper, we have investigated the target object information collection problem in multi-tagged COTS RFID systems with faulty tags. First, we have presented an efficient and compact tag pseudo-ID design, which enables the reader to select any combination of tags in a bucket with only one SELECT. Second, we have developed a fault-handling mechanism that can harness the first reply backscattered by a selected tag in a slot (without executing the entire slot) to check if the tag is faulty. Third, armed with the above two techniques, we have designed a novel COTS RFID object information collection algorithm by leveraging the functionality specified in C1G2 to optimize the information collection sequence. Comparative experiments on an RFID system prototype have validated that our algorithm outperforms the best state-of-the-art solution by an average of

35-50%. Regarding further work, we plan to use our algorithmic framework to devise more C1G2-compliant RFID algorithms for multi-tagged COTS RFID systems.

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